CS6910 — Fundamentals of Deep Learning

Tier	A
Phase (order)	6
Do this first	CS4011, calculus
Estimated effort	12 weeks, ~10 h/week
Unlocks	CS6370, CS6350, CS7016, CS6886

Roadmap

1. Perceptrons, MLP, backpropagation, optimisers (SGD, Adam).
2. CNNs, RNNs/LSTMs; then attention and transformers.
3. Regularisation, normalisation, training recipes.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Deep Learning (100+ videos). NPTEL *Deep Learning* (Mitesh Khapra, IIT Madras): this is CS6910 itself.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Deep Learning (68 videos). IIT Madras BS *Deep Learning*: newer recording, includes transformers.
- Neural Networks: Zero to Hero (10 videos). Andrej Karpathy *Neural Networks: Zero to Hero*: build backprop, an MLP and GPT from scratch in code. Do it alongside Khapra.
- MIT 6.S191: Introduction to Deep Learning (90 videos). MIT 6.S191 *Introduction to Deep Learning* (updated every year): fast overview + labs.

⚡ Short videos for intuition and revision

- Neural networks (10 videos). 3Blue1Brown *Neural networks*: backprop + transformers visualised. Watch first.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–18	History, MP neuron, perceptron, convergence proof. Code a perceptron.
2	19–34	Sigmoid neurons, gradient descent, feedforward nets, backpropagation. Write backprop for an MLP in NumPy and check with numerical gradients.

Week	Lectures	What to do
3	35–43	Optimisers: momentum, NAG, SGD/mini-batch, learning-rate schedules, Adagrad/RMSProp/Adam. Compare them on MNIST.
4	44–51, 33	Linear algebra recap, PCA, SVD.
5	52–57	Autoencoders (denoising, sparse, contractive). Train one in PyTorch.
6	58–69	Bias-variance, regularisation, augmentation, early stopping, dropout.
7	70–74	Pre-training, activations, initialisation, batch norm.
8	75–86	Word representations: word2vec, GloVe.
9	87–100	CNNs and visualising them. Train a CNN on CIFAR-10.
10-12	rest of playlist (after 100)	RNNs, LSTMs, seq2seq, attention, transformers: open the full playlist. Then Karpathy's GPT video (alt).

▼ **All lectures in *Deep Learning* (100+ videos)**

1. Deep Learning - Course Introduction
2. Deep Learning(CS7015): Lec 1.1 Biological Neuron
3. Deep Learning(CS7015): Lec 1.2 From Spring to Winter of AI
4. Deep Learning(CS7015): Lec 1.3 The Deep Revival
5. Deep Learning(CS7015): Lec 1.4 From Cats to Convolutional Neural Networks
6. Deep Learning(CS7015): Lec 1.5 Faster, higher, stronger
7. Deep Learning(CS7015): Lec 1.6 The Curious Case of Sequences
8. Deep Learning(CS7015): Lec 1.7 Beating humans at their own games (literally)
9. Deep Learning(CS7015): Lec 1.8 The Madness (2013-)
10. Deep Learning(CS7015): Lec 1.9 (Need for) Sanity
11. Deep Learning(CS7015): Lec 2.1 Motivation from Biological Neurons
12. Deep Learning(CS7015): Lec 2.2 McCulloch Pitts Neuron, Thresholding Logic
13. Deep Learning(CS7015): Lec 2.3 Perceptrons
14. Deep Learning(CS7015): Lec 2.4 Error and Error Surfaces
15. Deep Learning(CS7015): Lec 2.5 Perceptron Learning Algorithm
16. Deep Learning(CS7015): Lec 2.6 Proof of Convergence of Perceptron Learning Algorithm
17. Deep Learning(CS7015): Lec 2.7 Linearly Separable Boolean Functions
18. Deep Learning(CS7015): Lec 2.8 Representation Power of a Network of Perceptrons
19. Deep Learning(CS7015): Lec 3.1 Sigmoid Neuron
20. Deep Learning(CS7015): Lec 3.2 A typical Supervised Machine Learning Setup
21. Deep Learning(CS7015): Lec 3.3 Learning Parameters: (Infeasible) guess work
22. Deep Learning(CS7015): Lec 3.4 Learning Parameters: Gradient Descent
23. Deep Learning(CS7015): Lec 3.5 Representation Power of Multilayer Network of Sigmoid Neurons
24. Deep Learning(CS7015): Lec 4.1 Feedforward Neural Networks (a.k.a multilayered network of neurons)
25. Deep Learning(CS7015): Lec 4.2 Learning Paramters of Feedforward Neural Networks (Intuition)

26. Deep Learning(CS7015): Lec 4.3 Output functions and Loss functions
27. Deep Learning(CS7015): Lec 4.4 Backpropagation (Intuition)
28. Deep Learning(CS7015): Lec 4.5 Backpropagation: Computing Gradients w.r.t. the Output Units
29. Deep Learning(CS7015): Lec 4.6 Backpropagation: Computing Gradients w.r.t. Hidden Units
30. Deep Learning(CS7015): Lec 4.7 Backpropagation: Computing Gradients w.r.t. Parameters
31. Deep Learning(CS7015): Lec 4.8 Backpropagation: Pseudo code
32. Deep Learning(CS7015): Lec 4.9 Derivative of the activation function
33. Deep Learning(CS7015): Lec 6.6 PCA : Interpretation 3
34. Deep Learning(CS7015): Lec 4.10 Information content, Entropy & cross entropy
35. Deep Learning(CS7015): Lec 5.1 & Lec 5.2 Recap: Learning Parameters: Guess Work, Gradient Descent
36. Deep Learning(CS7015): Lec 5.3 Contours Maps
37. Deep Learning(CS7015): Lec 5.4 Momentum based Gradient Descent
38. Deep Learning(CS7015): Lec 5.5 Nesterov Accelerated Gradient Descent
39. Deep Learning(CS7015): Lec 5.6 Stochastic And Mini-Batch Gradient Descent
40. Deep Learning(CS7015): Lec 5.7 Tips for Adjusting Learning Rate and Momentum
41. Deep Learning(CS7015): Lec 5.8 Line Search
42. Deep Learning(CS7015): Lec 5.9 Gradient Descent with Adaptive Learning Rate
43. Deep Learning(CS7015): Lec 5.9 (Part-2) Bias Correction in Adam
44. Deep Learning(CS7015): Lec 6.1 Eigenvalues and Eigenvectors
45. Deep Learning(CS7015): Lec 6.2 Linear Algebra : Basic Definitions
46. Deep Learning(CS7015): Lec 6.3 Eigenvalue Decompositon
47. Deep Learning(CS7015): Lec 6.4 Principal Component Analysis and its Interpretations
48. Deep Learning(CS7015): Lec 6.5 PCA : Interpretation 2
49. Deep Learning(CS7015): Lec 6.6 (Part-2) PCA : Interpretation 3 (Contd.)
50. Deep Learning(CS7015): Lec 6.7 PCA : Practical Example
51. Deep Learning(CS7015): Lec 6.8 Singular Value Decomposition
52. Deep Learning(CS7015): Lec 7.1 Introduction to Autoncoders
53. Deep Learning(CS7015): Lec 7.2 Link between PCA and Autoencoders
54. Deep Learning(CS7015): Lec 7.3 Regularization in autoencoders (Motivation)
55. Deep Learning(CS7015): Lec 7.4 Denoising Autoencoders
56. Deep Learning(CS7015): Lec 7.5 Sparse Autoencoders
57. Deep Learning(CS7015): Lec 7.6 Contractive Autoencoders
58. Deep Learning(CS7015): Lec 8.1 Bias and Variance
59. Deep Learning(CS7015): Lec 8.2 Train error vs Test error
60. Deep Learning(CS7015): Lec 8.2 (Part-2) Train error vs Test error (Recap)
61. Deep Learning(CS7015): Lec 8.3 True error and Model complexity
62. Deep Learning(CS7015): Lec 8.4 L2 regularization
63. Deep Learning(CS7015): Lec 8.5 Dataset augmentation
64. Deep Learning(CS7015): Lec 8.6 Parameter sharing and tying
65. Deep Learning(CS7015): Lec 8.7 Adding Noise to the inputs

66. Deep Learning(CS7015): Lec 8.8 Adding Noise to the outputs
67. Deep Learning(CS7015): Lec 8.9 Early stopping
68. Deep Learning(CS7015): Lec 8.10 Ensemble Methods
69. Deep Learning(CS7015): Lec 8.11 Dropout
70. Deep Learning(CS7015): Lec 9.1 A quick recap of training deep neural networks
71. Deep Learning(CS7015): Lec 9.2 Unsupervised pre-training
72. Deep Learning(CS7015): Lec 9.3 Better activation functions
73. Deep Learning(CS7015): Lec 9.4 Better initialization strategies
74. Deep Learning(CS7015): Lec 9.5 Batch Normalization
75. Deep Learning(CS7015): Lec 10.1 One-hot representations of words
76. Deep Learning(CS7015): Lec 10.2 Distributed Representations of words
77. Deep Learning(CS7015): Lec 10.3 SVD for learning word representations
78. Deep Learning(CS7015): Lec 10.3 (Part-2) SVD for learning word representations (Contd.)
79. Deep Learning(CS7015): Lec 10.4 Continuous bag of words model
80. Deep Learning(CS7015): Lec 10.5 Skip-gram model
81. Deep Learning(CS7015): Lec 10.5 (Part-2) Skip-gram model (Contd.)
82. Deep Learning(CS7015): Lec 10.6 Contrastive estimation
83. Deep Learning(CS7015): Lec 10.7 Hierarchical softmax
84. Deep Learning(CS7015): Lec 10.8 GloVe representations
85. Deep Learning(CS7015): Lec 10.9 Evaluating word representations
86. Deep Learning(CS7015): Lec 10.10 Relation between SVD and Word2Vec
87. Deep Learning(CS7015): Lec 11.1 The convolution operation
88. Deep Learning(CS7015): Lec 11.2 Relation between input size, output size and filter size
89. Deep Learning(CS7015): Lec 11.3 Convolutional Neural Networks
90. Deep Learning(CS7015): Lec 11.3 (Part-2) Convolutional Neural Networks (Contd.)
91. Deep Learning(CS7015): Lec 11.4 CNNs (success stories on ImageNet)
92. Deep Learning(CS7015): Lec 11.4 (Par-2) CNNs (success stories on ImageNet) (Contd.)
93. Deep Learning(CS7015): Lec 11.5 Image Classification continued (GoogLeNet and ResNet)
94. Deep Learning(CS7015): Lec 12.1 Visualizing patches which maximally activate a neuron
95. Deep Learning(CS7015): Lec 12.2 Visualizing filters of a CNN
96. Deep Learning(CS7015): Lec 12.3 Occlusion experiments
97. Deep Learning(CS7015): Lec 12.4 Finding influence of input pixels using backpropagation
98. Deep Learning(CS7015): Lec 12.5 Guided Backpropagation
99. Deep Learning(CS7015): Lec 12.6 Optimization over images
100. Deep Learning(CS7015): Lec 12.7 Create images from embeddings

Only the first 100 are listed; see the full playlist.

Books, notes and websites

- ✅ NPTEL *Deep Learning* (Mitesh Khapra, IIT Madras) — nptel.ac.in/courses/106106201/
- d2l.ai *Dive into Deep Learning* (free, with code)

- Karpathy, *Neural Networks: Zero to Hero* (YouTube)
- Stanford CS231n; fast.ai Part 1
- Book: Goodfellow et al., *Deep Learning* — deeplearningbook.org

Practice / proof of understanding

Write backprop for an MLP in NumPy, train a CNN on CIFAR-10 in PyTorch, and train a tiny GPT on a small text file.

What your IITM course sheet says

From the PDF: CS6910

Official description: Neural Networks have made tremendous impact on various AI fields in the recent past. In this course, we study the basics of Neural Networks and their various variants such as the Convolutional Neural Networks and Recurrent Neural Networks. We also study the different ways in which they can be used to solve problems in various domains such as Computer Vision, Speech and NLP. We also look at the latest results and trends in the field.

Official textbooks: none

Official references: 1. Deep Learning, An MIT Press book, Ian Goodfellow and Yoshua Bengio and Aaron Courvill. 2. InformationTheory,Inference,andLearningAlgorithms(Ch.5),DavidMacKay. 3. Latest research papers from various Computer Vision, Natural Language Processing, Speech and Information Retrieval conferences.